Examples of Systems Studied by Network Scientists — Part 1

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Here's an example of a transportation network. So in the urban and the rural interstate
highways, and other major highways in the US, the nodes in such a network could be all
junctions or entry, exit points on a highway, while the edges can be the highway segments
between such points. Note that the network of this type would typically have weighted
edges, meaning that each edge would be associated with a number showing its capacity. For
example, the number of lanes at that highway segment. Here is another transportation
network example, this time for the direct flights of delta airlines in North America.
This network would also be weighted, even though the weights are not shown in this
visualization, with the weight of each edge being the number of available seats in all
flights between the corresponding two airports on a daily basis. You can actually find
such data for every airline through the Bureau of Transportation Statistics, which is
part of the US Department of Transportation. See the link below this video. Here is an
example of a computer network used in many data centers, referred to as the Fat Tree
Topology. An important difference in this kind of graph representation is that we need to
consider that not all nodes are of the same type. Some are servers, some are access
switches, aggregation switches, our core switches. This difference means that each node
in this graph is associated with a different computation or communication capacity. We
should be careful to not ignore such heterogeneity in the nature of nodes or edges.
Otherwise, it is possible to reach completely wrong conclusions. Let's move now from
technological to human or social networks. In these networks, the nodes are typically
specific individuals, and the edges represent different types of relation or contact. For
instance, this figure is based on data from a very interesting study at the high school.
800 students were asked whether they had sexual relations during the past six months with
another student at the same school. The network shows only those individuals that had at

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least one such relation in the given time period. Note that the network in this case does
not consist of a connected graph. There are several connected components, with one of
them at the top left being much larger than the others. This is common in most real-world
networks. There may be more than one connected components, but typically, one of them
covers the majority of the nodes. Note that most nodes have a small number of
connections. For example, there are 63 pairs of nodes that do not connect to anyone else.
But there are also few nodes with a much larger number of connections. This heterogeneity
is a common property of many real-world networks. Online social networks have transformed
our lives in the last 15 years or so. Here you see a small portion of the Twitter
network, showing the connections between a number of poker players. In Twitter, a user
can choose to follow another user. This means that the edges in the network are directed
from one node to another. Of course, we can also have bi-directional edges, when two
people follow each other. By the way, social networking is not a human invention. A large
number of species create complex and dynamic social networks. This network refers to 62
dolphins that were monitored for a period of about seven years. Two individuals are
connected if they spent significant amount of time swimming together. The researchers
that first performed this study noticed two strongly formed communities, the blue and the
red. And a more weakly formed community, the green, that often interacts with both of the
blue and red communities. The dolphins shown as black dots were not included in the
original data analysis. The blue dashed line represents how a community detection
algorithm would separate the dolphins into two groups without any information about the
color of each node. Note that the algorithm is doing a pretty good job separating the
blue and red communities, and it also suggests what group the black nodes most likely
belong to. Networks sometimes refer to the exchange of products, materials, or just
money. Such networks are viable in the field of economics and finance. The networks here
show three different snapshots from 1995 to 2011, showing the trade volume between
different countries, only the countries with the largest trade values. The visualization
does not show numbers, but the edges are in fact directed and weighted showing the total
trade volume from country x to country y during that year. This is a good example of a
dynamic network, a network the changes over time. Note how much denser the network has

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gradually become. You can also see many other interesting changes, such as the major
shift of China from a peripheral to a central role in the world trade network. If you
want to analyze this data yourself, they are publicly available at the world trade
organization link shown below.

Examples of Systems Studied by Network Scientists — Part 2

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Let's now switch to biological systems. There are several mechanisms in a living cell
that depend on network interactions rather than individual chemical reactions or
molecules. Most of the functions of a biological cell are performed through proteins. You
can think of proteins as Lego pieces of specific shape and size that can interact with
other proteins to perform a certain function, such as signal transduction, transport of
other proteins or DNA transcription. The set of all possible interactions between protein
pairs in a cell forms the protein-protein interaction, or PPI network of that cell. This
network is extremely important, not only in understanding how the cell performs its
function, but also to understand diseases in which one or more proteins does not fold
properly, such as Alzheimer's, ALS or sickle cell disease. Additionally, PPI networks are
instrumental in designing drugs that target specific viruses or other pathogens. This
figure shows the PPI network of a single cell organism called fission yeast, and it
consists of about 3,400 proteins and the 37,000 in directions. The different colored
clusters in the network correspond to specific biological functions. As you can see, most
functions are performed by groups of highly interconnected proteins. But there are also
many proteins that bridge those clusters, contributing in more than one biological
function. Ecologists also rely on networks to represent the interactions between species
in an ecosystem. These interactions can take many forms, but in the simplest case we can
consider only trophic links. In other words, who is eating what? The network in this
visualization shows with the plants as green nodes in the center and crops in light

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green. It's type of consumer species at the periphery of the circle has a unique color.
Note that some of the consumers eat both plants and animals. Such networks are crucial in
predicting the effect of human interventions, herbicides, etc. The brain is also a system
that is actively studied as a network, both at the level of individual neurons and at the
lower spatial resolution at the level of brain regions. This figure shows a sketch of two
neurons that connect through a synapses. When the left neuron, called pre-synaptic, fires
it generates an action potential, in other words. The receiving neuron at the right will
get depolarized and may also generate an action potential. Note that the connection is
directed and weighted by the number and strength of the corresponding synapses. Our brain
is nothing but a huge network of interconnected neurons. The human brain includes about
86 billion neurons and more than 100 trillion synapses, a huge network indeed. But it is
important to pause for a moment and consider that all our impressive capabilities and
fallacies are generated not by the complexity of individual neurons, but by the
complexity of the network that interconnects those neurons. The neurons in the human
brain are practically the same cells as the neurons in any other animal. What makes us
humans, rather than flies or cats, is the network within our skull. As mentioned earlier
in this lesson, we know the complete neural network of a few species, such as the
microscopic worm, C-elegans. This visualization shows neurons with three different
colors. Red for sensory neurons providing input to a neural system. Green for motor
neurons generating output action by connecting to muscles. And blue for interneurons
computing the intermediate level functions. It is extremely unlikely that we will have
similar maps for the entire human brain in the near future. Instead, the human brain
network is usually started at the courser level of spacial resolution using magnetic
resonance imaging, or MRI, which is a non-invasive neuro imaging technology. Diffusion
weighted MRI provides maps showing how different brain regions, down to a voxel dimension
of about one millimeter, are connected to each other. These anatomical connections can be
inferred using some very interesting algorithms that perform tractography and they allow
us to create structural brain networks, as shown in the upper part of the figure.
Functional MRI, or fMRI, on the other hand, provides a time series for its brain region
showing how active that region is as a function of time. This depends, of course, on the

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cognitive task that the subject performs during the MRI scan. The spatio temporal
correlations between different brain regions allows us to construct functional brain
regions, as shown at the lower part of the figure. We will discuss in more technical
detail how to infer and analyze functional networks derived from spatiotemporal data
later in the course.